

READ FIRST This brief is set in a custom typeface. Read it on screen or on paper. Text copied out of this file does not reproduce faithfully, so work from what you can see rather than from a paste.

THE CLIENT

Bright Path is a tutoring centre in Da Nang. It has 12 tutors on the payroll and a little under two hundred families on the books. Lessons are one-to-one, sixty or ninety minutes long, and they take place in six rooms above a bakery.

Everything is run from one shared spreadsheet and a set of WhatsApp groups. Mai, the receptionist, opens the spreadsheet at seven each morning, works out who is teaching whom, and messages each tutor their day. When a family cancels she edits the sheet, messages the tutor, and tries to remember to tell the tutor's next family that a slot has opened up. She is the only person who fully understands the sheet, and she goes on leave in eight weeks.

The owner has asked for an internal tool. There is no budget for a long build, so what you produce has to be the smallest thing that takes the daily pain away.

WHAT PEOPLE TOLD US

OWNER

“Twice last term we had a student booked into two places at once. The family turned up and nobody knew. That can never happen again — if the system allows it, the system is broken.”

“And I want to open the laptop and see today. Not scroll. See it.”

RECEPTIONIST

“Cancellations are the worst part. Someone messages at night, I fix the sheet in the morning, and by then the tutor has already left home.”

“In exam season I put two students with one tutor in the same room and the same slot, on purpose. It is half price and the families like it. I do that most weeks.”

TUTOR

“I get my day in a message, then a correction, then sometimes a third message. I do not always know which one is real.”

“I have driven in for a lesson that was called off the night before. Twice.”

HOW THE CENTRE ACTUALLY OPERATES

RULE	WHAT IT MEANS IN PRACTICE
Opening days	The centre teaches Tuesday to Sunday. It is closed on Monday, when the rooms are cleaned. Lessons run from mid-morning to mid-evening.
Tutor load	No tutor may be given more than 6 bookings in a single day. Mai breaks this rule when she is desperate, and the owner wants it enforced.
Late cancellation	A family may cancel free of charge up to 4 hours before the lesson starts. Inside that window the lesson is charged in full and the tutor is still paid. Cancelling frees the room and the slot; a student who simply does not arrive is a no show, which frees neither.
The cut-off	Tomorrow's schedule is final at 16:00 today. Changes after the cut-off still happen, but they must be visible as changes rather than quietly overwriting what the tutor was already told.
Rooms	Six rooms. A room holds one lesson at a time, whoever is teaching in it, and a tutor can only be in one room at a time.

You are given an export of the real spreadsheet, lightly anonymised and tidied so that it loads without a fight. Every row parses and every name resolves. What the rows describe is another matter: they are what the centre actually ran that week, which is not always what the rules above say should have happened.

YOUR ASSIGNMENT

Four phases. What you choose to build, and why, weighs as much with us as the code: we would rather read one feature you can defend than four you cannot.

PHASE	WHAT TO PRODUCE
1 · Read the situation	Open a file named DECISIONS.md at the root of your repository. In it: the questions you would put to the owner first, and how each answer would change what you build; anything in this brief that does not hold together, with the reading you chose; and the assumptions you had to invent.
2 · Choose what to build	List the features you can see this tool needing — one line each, half a dozen is plenty. Then pick exactly one and argue for it: why is that one worth more to Bright Path than the others, in the time you have? Say what you leave broken by choosing it.
3 · Design and build that one	In the same file: the part of your data model that feature needs, and how you represent a booking cancelled or moved after the tutor was told; which rules above you enforce in the database and which in code, and why; its API shape, plus one endpoint you rejected. Then build it, and only it — any language, framework and database, using the stack you are quickest in. Load the export as seed data, and put the commands to run it in your README with what you saw.
4 · Reflect	Close DECISIONS.md honestly: what you would build next with another week, what you know is weak, where your AI assistant helped, and one suggestion you threw away — and why you were right to.

HOW WE WILL READ YOUR SUBMISSION

WHAT WE WEIGH	WHAT EARNS MARKS
Problem reading	You worked out what actually hurts, told it apart from what was merely mentioned, and noticed where the brief argues with itself.
Prioritisation	You picked the feature that earns its place first, and can say why the rest wait. One feature that works beats three that half-work.
Data model	The schema makes the centre's rules hard to break and survives a booking changing after the fact.
API and code	It runs. Boundaries are sensible, names are honest, and somebody else could pick this up on Monday.
Communication and reflection	You wrote for a busy reader who was not in the room, and your reflection is honest about what is weak.

PRACTICAL DETAILS – NOT PART OF THE PUZZLE

- **Time box: two and a half hours, and please do not exceed it.** One feature you can explain is exactly what we want. A few bullets per written section is plenty.
- **Pin your dates.** The seed data covers one particular week. Where your tool needs to know what today is, use a date inside that week rather than the real clock, and say which one in your write-up.
- **Send us a link to a GitHub repository** – public, or private with an invitation. Push the whole history and keep your commits atomic: one commit per change, with a message saying what and why. We read them, so please do not squash before sending.
- **AI assistants are encouraged** – whatever you use daily. We assume you used one and will ask about it, so please do not pretend otherwise.
- **We will go through your solution with you at the technical interview.** Expect to explain any part of it, and to change something while we watch. Submit only code you understand and could have written yourself – owning what you send is the point.

Bright Path is a composite of real clients; the data is synthetic. Questions about the brief are welcome at any point – asking a good one is part of the job.